FLAT BASE CHANGE PRESERVES PSEUDO-COHERENT MORPHISMS OF SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17387-17390 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17392-17399, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE: pure 'This follows from the more general Lemma <exact>' pointer; flagged for the Chief.

Editorial note. Counted A side only (spaces twin proof incomplete per ruling v3). Cited inside L16244's proof.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, lemma base-change-relative-pseudo-coherent.

Source reference. Original lemma 067A: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17399–17399. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A flat base change of a pseudo-coherent morphism is pseudo-coherent.*

Proof. This translates into the following algebra result: Let $A \to B$ be a pseudo-coherent ring map. Let $A \to A'$ be flat. Then $A' \to B \otimes_A A'$ is pseudo-coherent. This follows from the more general More on Algebra, Lemma 067A. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.